

Design Calculation Sheet — VBF-T4R1-CALC-01

Quantity	Basis	Value	Unit	Source
1. Input parameters				
Current collectors / separator / particle / transport / cooling overrides	design overrides (bridge/params_r6_t2.json)	see bridge file	-	9 keys: CC 10/8 um, sep 10 um, R_neg 2.5 um, t+ 0.6, kappa 1.1, D_e 3e-10, h 80
Positive electrode thickness	parameter set	75.6	um	Chen2020
Negative electrode thickness	parameter set	85.2	um	Chen2020
Separator thickness	override	10.0	um	bridge/params_r6_t2.json
Positive electrode porosity	parameter set	0.335	-	Chen2020
Negative electrode porosity	parameter set	0.25	-	Chen2020
Separator porosity	parameter set	0.47	-	Chen2020
Positive electrode density	parameter set	3262	kg/m3	Chen2020
Negative electrode density	parameter set	1657	kg/m3	Chen2020
Electrode height x width	parameter set	0.065 x 1.58	m	Chen2020 -> area 0.1027 m2
Voltage window	parameter set	2.5 - 4.2	V	Chen2020 cut-offs
Nominal cell capacity	parameter set	5.0	Ah	Chen2020
2. Capacity and energy				
1C discharge capacity (25 C)	run-pyamm 1C_discharge DFN	5.03677	Ah	cell/r6_t2_1c_dfn.json:capacity_ah
-20 C discharge capacity (cold-soak)	run-pyamm lowT_discharge DFN, T_init 253.15 K, h=80	4.99987	Ah	cell/r6_t2_lowT_dfn.json:capacity_ah
-20 C retention	capacity(lowT)/capacity(1C)	0.99267	-	cell/r6_t2_retention.json:lowT_retention
Discharge energy	time integration of V*I over 1C discharge	17.9505	Wh	cell/r6_t2_energy.json:energy_wh
Midpoint voltage	calc-energy	3.742	V	cell/r6_t2_energy.json:midpoint_voltage_v
DC resistance	calc-energy DCR caliber	0.004065	ohm	cell/r6_t2_energy.json:dcr_ohm
3. Energy density				
Cell mass (contract)	sum layer thickness x (1-porosity) x density x area, electrolyte excluded	0.0380668	kg	cell/r6_t2_energy.json:mass_kg
Cell volume	sum layer thickness x area = 188.8e-6 x 0.1027	1.938976e-5	m3	cell/r6_t2_energy.json:volume_m3
Energy density	energy / mass	471.55	Wh/kg	cell/r6_t2_energy.json:energy_density_wh_kg
Volumetric energy density	energy / volume	925.77	Wh/L	cell/r6_t2_energy.json:energy_density_wh_l
Power density	calc-energy	26903.6	W/kg	cell/r6_t2_energy.json:power_density_w_kg
4. NP ratio and mass				
N/P (total capacity density basis)	(eps_am_neg x c_max_neg x L_neg) / (eps_am_pos x c_max_pos x L_pos)	0.667	-	(0.75x33133x85.2)/(0.665x63104x75.6); parameter-set values
N/P operational balance	same 5.037 Ah window on both electrodes	~1.0	-	neg 90.1->3.7%, pos 27.0->84.7% (initial stoichiometries, measured window)
Positive electrode mass	layer_kg_m2 x area	16.842	g	cell/r6_t2_energy.json
Negative electrode mass	layer_kg_m2 x area	10.874	g	cell/r6_t2_energy.json
Al collector mass	layer_kg_m2 x area	2.773	g	cell/r6_t2_energy.json

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Cu collector mass	layer_kg_m2 x area	7.362	g	cell/r6_t2_energy.json
Separator mass	layer_kg_m2 x area	0.216	g	cell/r6_t2_energy.json
Total mass	sum	38.067	g	cell/r6_t2_energy.json:mas s_kg
5. Process parameters				
Areal density positive	$L \times (1-\epsilon) \times \rho$	163.99	g/m2	cell/r6_t2_energy.json:layer _kg_m2
Areal density negative	$L \times (1-\epsilon) \times \rho$	105.88	g/m2	cell/r6_t2_energy.json:layer _kg_m2
Compaction density positive	$\rho \times (1-\epsilon) / 1000$	2.169	g/cm3	parameter set (divide by 1000)
Compaction density negative	$\rho \times (1-\epsilon) / 1000$	1.243	g/cm3	parameter set (divide by 1000)
Electrolyte fill amount	pore volume x 1.2 g/cm3 (literature default)	6.325	g	pore volume 5.271 mL
Formation recommendation	0.1C CC charge to 4.2 V, 25 C, 2 cycles	-	-	design recommended value; production-line value requires tuning